

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal report presents the design of an AI-powered Emotion Detection & Learning Support Engine that identifies students' learning emotions from text and delivers personalized educational guidance. It combines a fine-tuned BERT model and a Multi-label BiLSTM network with Google Gemini AI to improve student engagement, reduce confusion, and provide emotion-aware learning support.

Project Overview	
Objective	The primary objective is to detect students' learning emotions from free-text learning problems and generate personalized, emotion-aware study guidance using deep learning and generative AI.
Scope	The project analyzes a student's selected subject and learning problem text to detect primary and mixed emotions, compares BERT and BiLSTM model predictions, generates AI-powered study guidance and study plans, and stores interaction history and analytics for future use.
Problem Statement	
Description	Many students experience confusion, frustration, or loss of motivation while learning, but traditional e-learning platforms provide the same experience to every learner without understanding their emotional state.
Impact	Solving this problem improves student engagement, reduces frustration, builds learning confidence, and increases the effectiveness of personalized educational support.
Proposed Solution	
Approach	Employ a fine-tuned BERT model and a Multi-label BiLSTM network to detect primary and mixed learning emotions, and use Google Gemini AI to generate personalized study guidance, delivered through an interactive Streamlit interface.
Key Features	- AI-based emotion detection using BERT and BiLSTM. - Mixed emotion detection and model comparison. - Personalized study guidance and study plan generation using Gemini AI. - Session history tracking and CSV storage. - Emotion analytics dashboard. - Interactive Streamlit interface. - Downloadable AI report.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Kaggle GPU (model training), Intel Core i5 Processor (local deployment)
Memory	RAM specifications	16 GB (training environment), 8 GB (local deployment)
Storage	Disk space for data, models, and logs	10 GB+ for datasets, model checkpoints, and session logs
Software		
Frameworks	Python frameworks	Streamlit, TensorFlow, Keras
Libraries	Additional libraries	Transformers, Torch, Pandas, NumPy, Plotly, Scikit-learn, Google Generative AI SDK
Development Environment	IDE	Kaggle Notebooks (training), Visual Studio Code (development)
Data		
Data	Source, size, format	Custom education-focused emotion dataset (~10,500+ samples), merged from public emotion datasets, stored in CSV format